Tech News

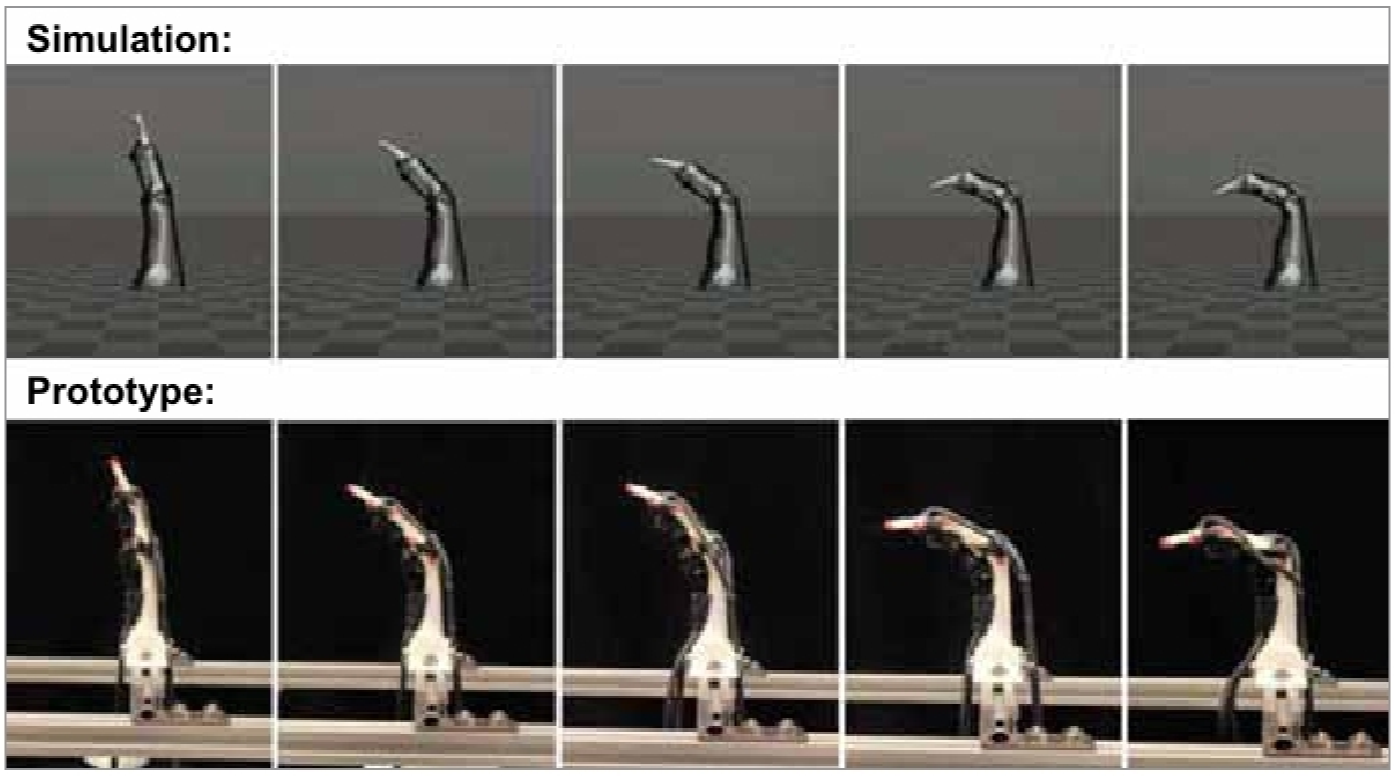
Technology Updates

Biomimetic robotic finger created using 3D printing

Researchers at the University of California, Santa Cruz and Ritsumeikan University, Japan, have designed and fabricated a robotic finger inspired by the human endoskeletal structure. This biomimetic robotic finger, presented at this year's International Conference on Ubiquitous Robots and Ambient Intelligence (URAI), was assembled using a multi-material 3D printer.

"Developing a robotic hand that has hard and soft components, just like the human hand, is a research topic that I wanted to explore for years," Maryam Tebyani, one of the researchers who carried out the study, told *TechXplore*.

The primary objective of the recent work by Tebyani and her colleagues was to create a robotic hand inspired by natural systems. However, the researchers wanted to focus most of their efforts on designing artificial hand rather than in fabricating and assembling it. They thus decided to take advantage of state-of-the-art 3D printing methods, which could simplify and accelerate the overall prototype production process.



Researchers have designed a robotic finger inspired by the human endoskeletal structure (Credit: Tebyani et al)

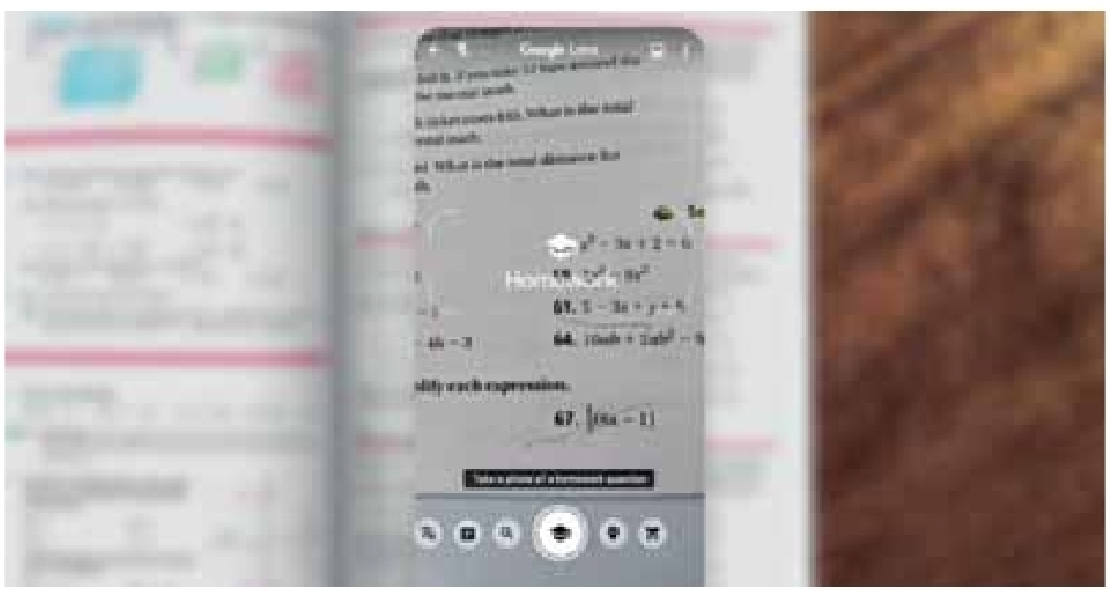
The robotic finger designed by the researchers has bone geometry, ligament structures, artificial muscles, and viscoelastic tendons that resemble those of humans. All of these components were synthesised as a single part using a multi-material 3D printer. The researchers also developed a model of the finger using a popular physics simulation engine called MuJoCo. This allowed them to compare the results achieved by their robotic finger in simulations with those achieved by a printed prototype of the finger in real-world settings.

Google Lens to assist students with homework

Google has announced that the company will soon be rolling out a new educational tool for Lens. Its purpose is to help students (and their home-teaching parents) solve and use math equations.

The Lens tool will be part of a suite of applications provided through Socratic, which Google bought last year. Currently, Socratic supports Math, Science, Literature, Social Studies, and several other subjects. Its purpose is to serve as an online tutor, helping students learn on their own.

To use the Lens tool, a student can take a picture of a formula or equation with their Android or iOS device. And then, the tool will provide quick access to helpful results such as step-by-step guides and detailed explainers that will assist in



Google to roll out a new educational tool for Lens (Credit: Google)